

SenNet  **BMS**

Building Automation



Solution components

- Datalogger running the Supervision and Control Framework
- 100% local execution (Edge Computing)



SenNet DL400 Datalogger



SenNet BMS Framework

SenNet DL400

Based on **Linux**, with high processing capacity thanks to its 4 cores.

Monitors energy consumption and generation and reports to the **EMS (Energy Management System)**.

- Integrates electrical meters through RS485 or wireless communication.
- Expandable digital inputs and outputs.
- Connectivity: Ethernet, SPE, Serial, WiFi, USB, and Radiofrequency.
- HDMI output for local synoptics.

Integrates a framework for building system control and supervision with Edge Computing operation.

Remote supervision and secure connection via Internet.

RF LongNet		433MHz@10mW / 869MHz (EU) - 915MHz (US) @25mW RX sensitivity: -124dBm
RF LoRaWAN		EU 868MHz TX power up to 27 dBm RX sensitivity down to -125dBm @ SF7, BW 125kHz
RF Zigbee		ISM Global 2.4 GHz TX power up to 20 dBm RX sensitivity down to -102dBm @ 250 kbps (ZC Coordinator Mode)
RF Wi-SUN		LPWAN (2.4 GHz / 863-870 / 915 MHz) RX sensibly down to -110dBm @ 50 kbps
M-Bus Wireless		868MHz (EU) – 915MHz (US) @25mW RX sensitivity -117dBm @ 4.8 kbps



CARACTERÍSTICAS GENERALES			
Power supply	8V...30Vdc (6W–10W depending on functionality and extensions)		
Connectivity	Ethernet 1Gbp/s 802.3bp	Single Port Ethernet 100Mbit/s 802.3bw	WiFi 5G / 2.4G (Station / Access Point Mode)
Access Port	Direct access via webserver IP:8080 (Datalogger webserver) IP:8085 (SenNet BMS webserver)		
Communications	RS485	RS232	RS232 (console)
	HDMI	USB 2.0	Bluetooth (console – optional)
Inputs/Outputs	8 digital inputs or outputs (Vpwr@100mA)		1 auxiliary power output (5V @300mA)
OS / Processor	Debian 12 Certified secure and anti-intrusion distribution		4-core ARM® Cortex®-A53 (1.4GHz)
Memory RAM / Disk 1 / Disk 2 (optional)	2GB RAM		16GB SSD (disk 1 - SO) 4GB SSD (Disk 2 - data logging)



TYPICAL INSTALLATION WITH ZIGBEE TECHNOLOGY



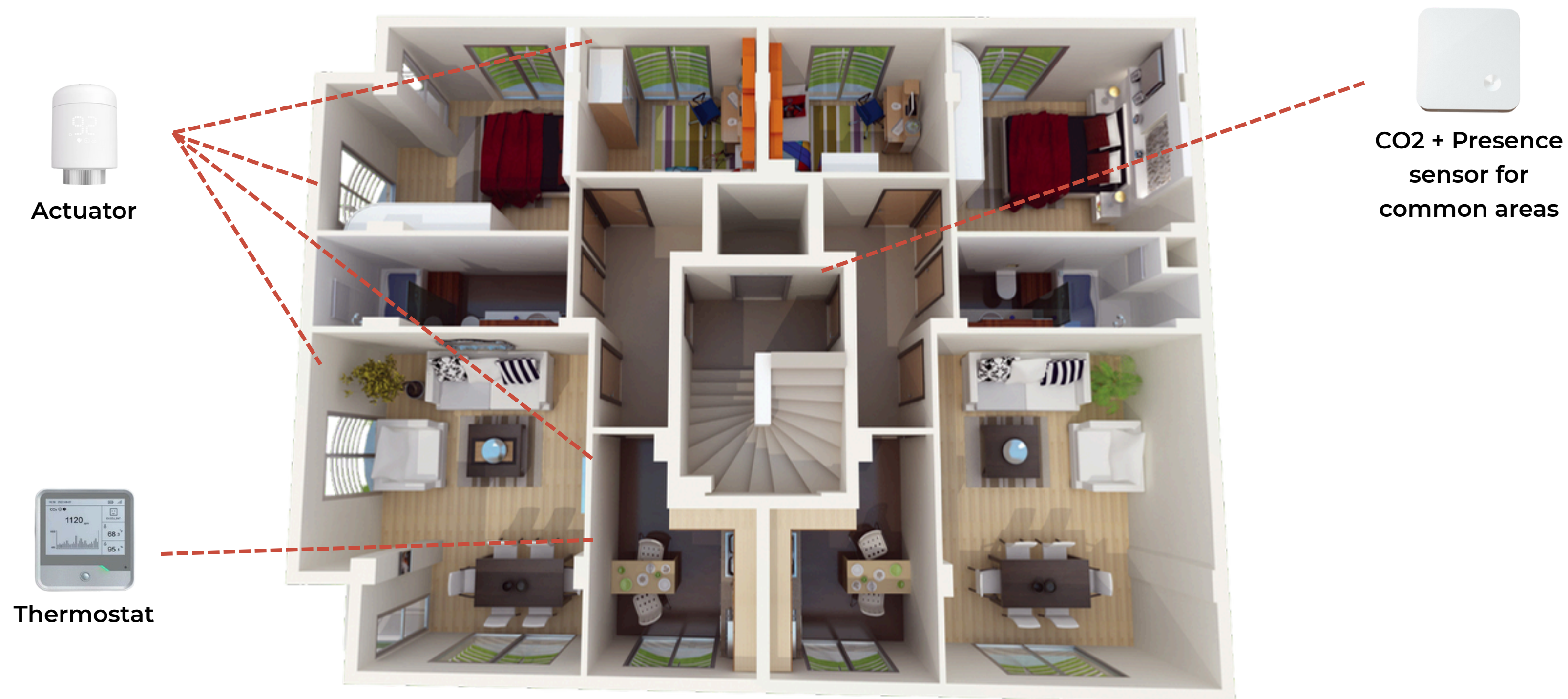
TYPICAL INSTALLATION WITH ZIGBEE TECHNOLOGY



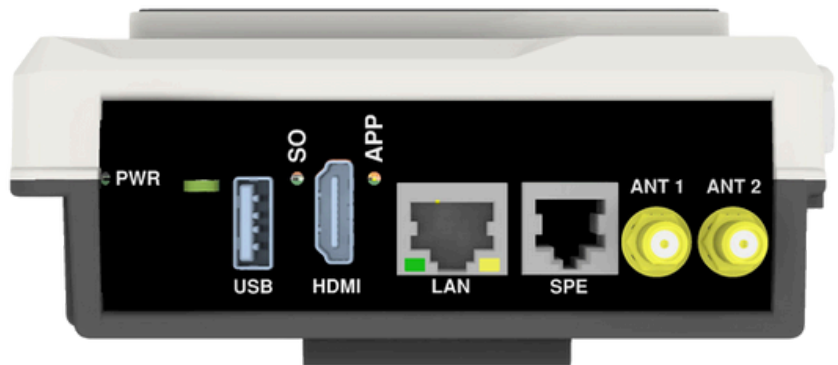
TYPICAL INSTALLATION WITH LORAWAN TECHNOLOGY



TYPICAL INSTALLATION WITH LORAWAN TECHNOLOGY

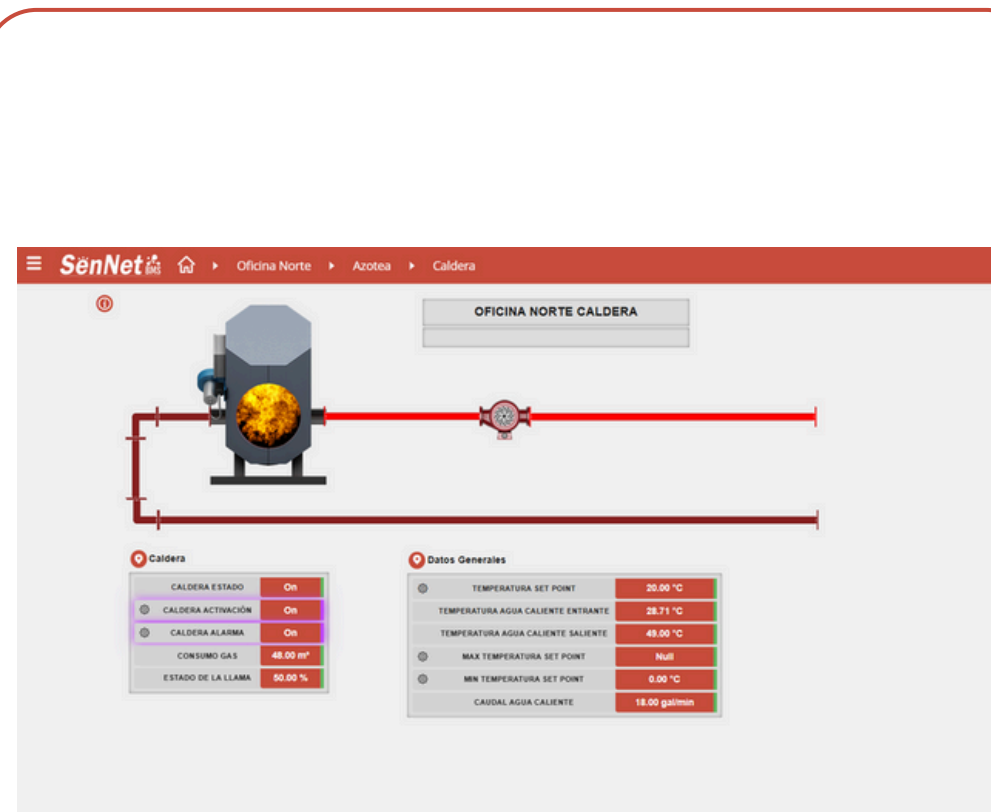


Available Controllers

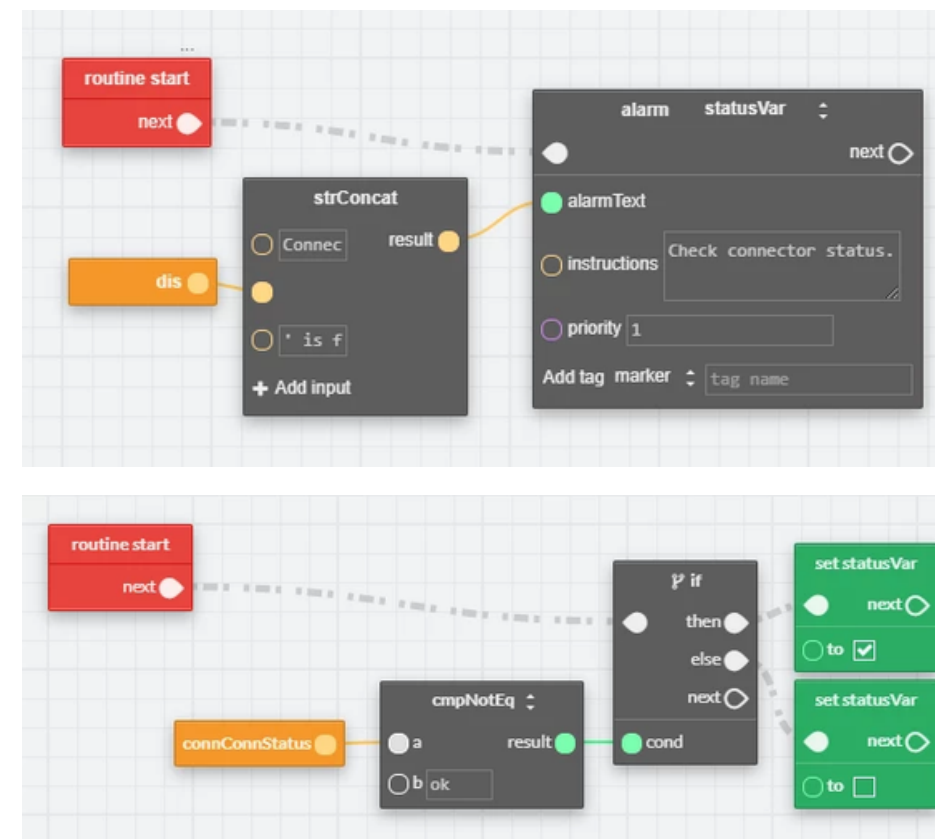


Modelo	Client WiFi / Access Point	Zigbee / W-Mbus / Wi-Sun *Default Zigbee, other options upon request		LoRaWAN	LongNet
DL400	☒	☒	☒	☐	☐
Módulo DL400 LoRaWAN	☒	☒	☒	☒	☐
Módulo DL400 LongNet	☒	☒	☒	☐	☒
Módulo DL400 LTE	☒	☒	☒	☐	☐

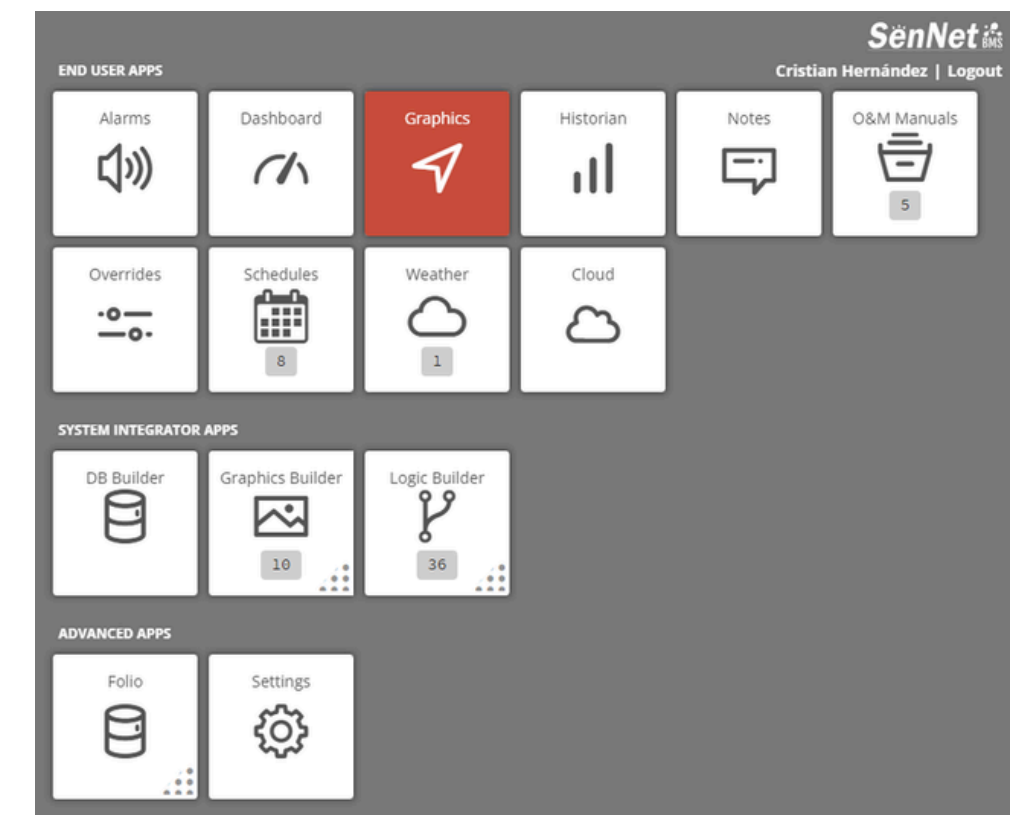
Integrated supervision and control within a powerful framework



**Customizable
interactive synoptics**

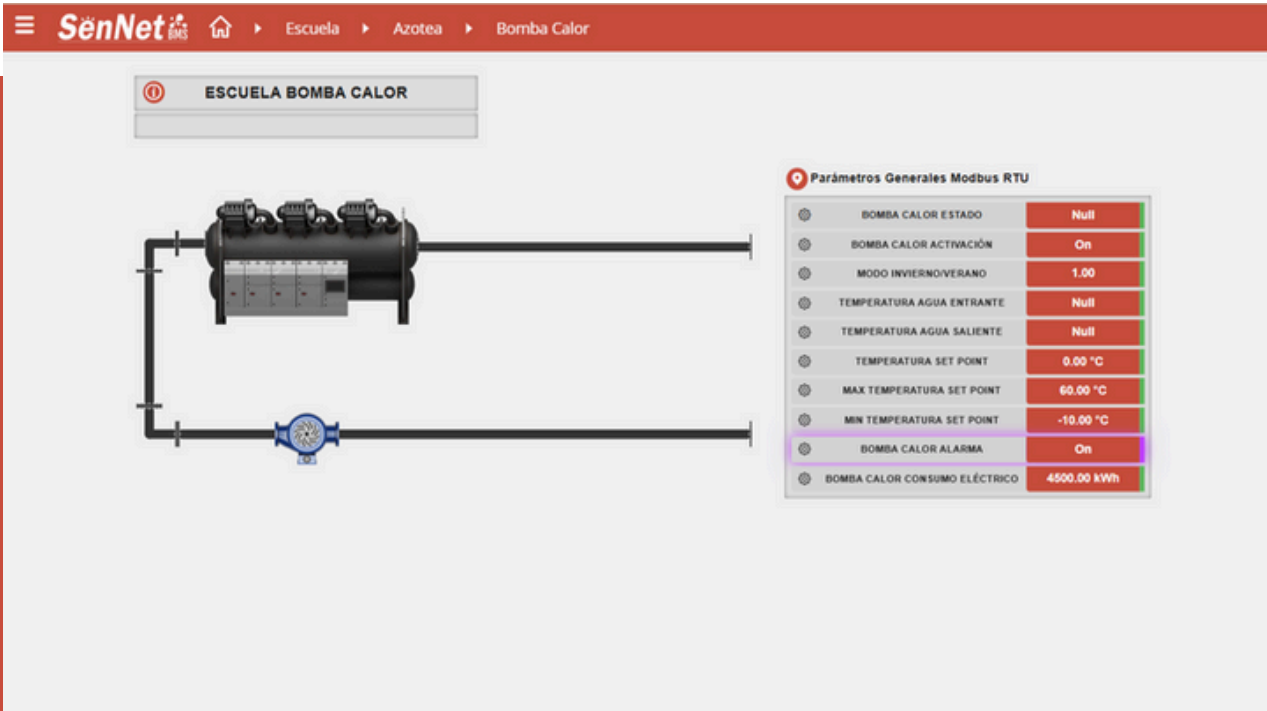
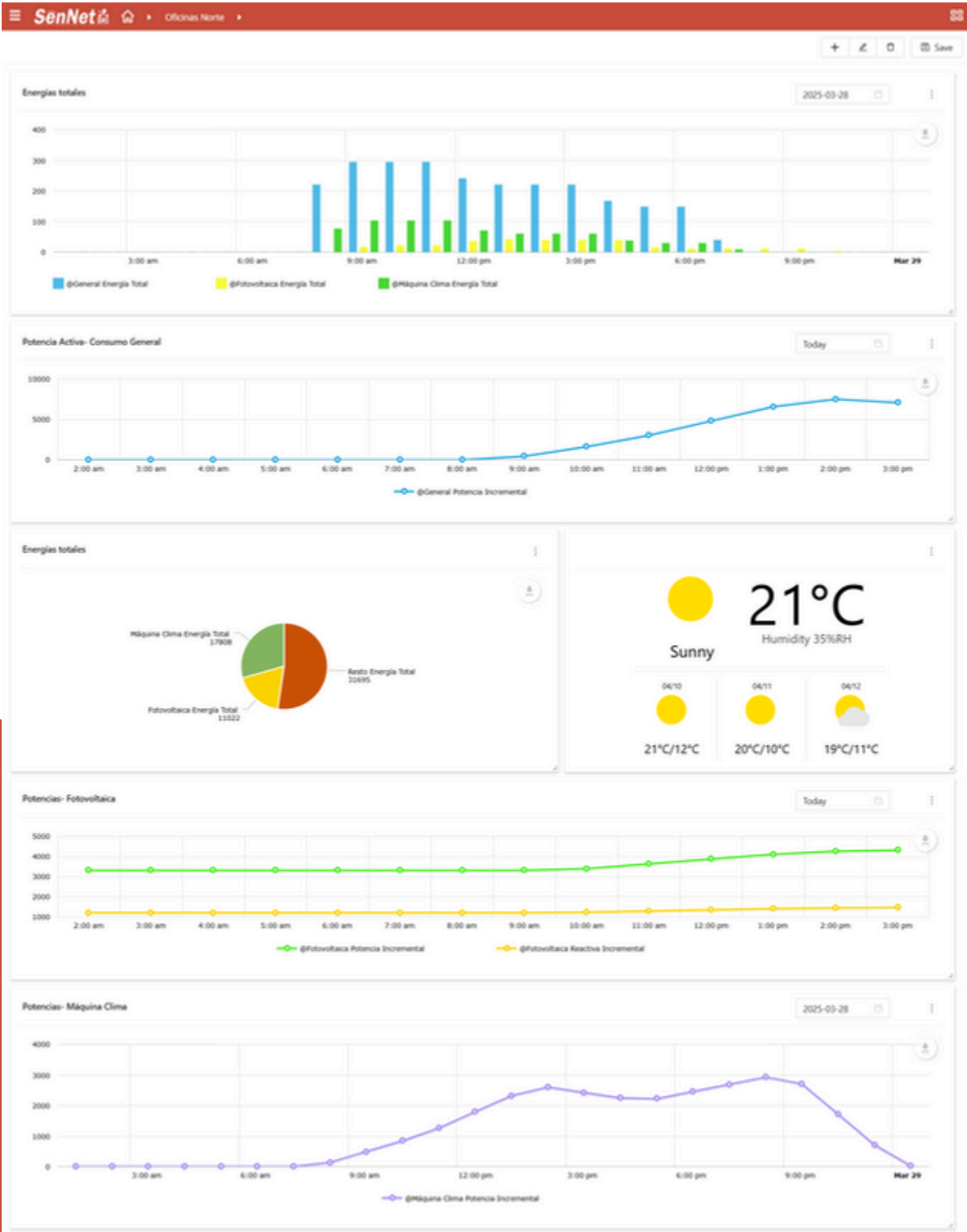


**Logic programming with
blocks or lines,** combining
simplicity and power.



**Includes a wide range of
applications for both users
and integrators.**

User Experience-oriented panels and graphs



Características del Framework

Configurable dashboards

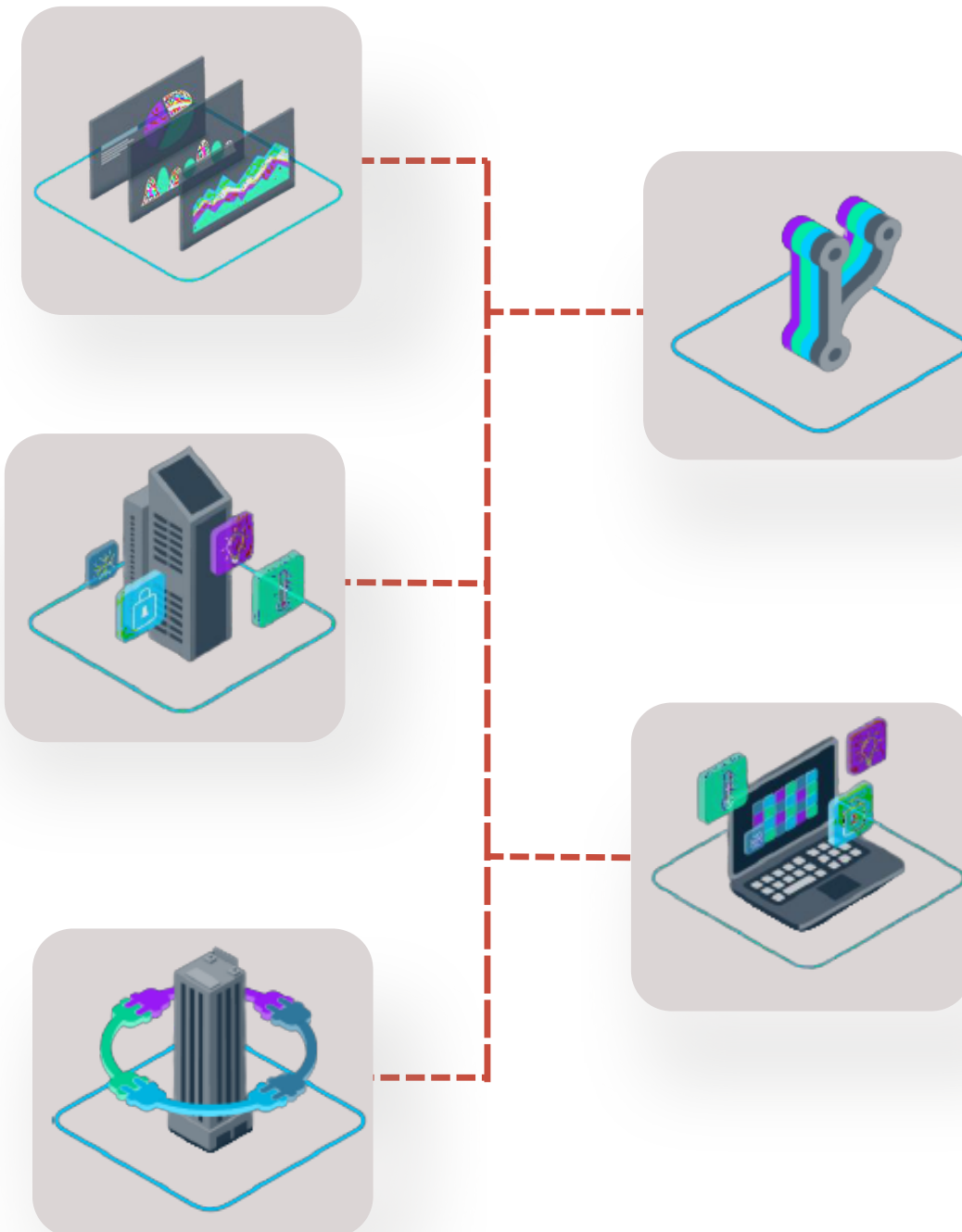
Easy-to-configure dashboards for advanced facility management.

Comprehensive management

Suitable for managing multiple different areas such as HVAC, lighting, shading, power, and energy management.

Device integration

Standard open protocols like BACnet, Modbus, Haystack, KNX, with advanced options (SQL, XML, REST, MQTT) and expandable to proprietary protocols.



State-of-the-art monitoring and control

Applications for alarms, scheduling, trends, 2D/3D graphs, reporting, and advanced control engine.

Workflow design

Tag-based workflow supported by assistants and templates, reducing engineering time by 50% compared to other building management systems.

Solution Scope



INTEGRATION OF MULTIPLE SYSTEMS

Compatible with all open protocols commonly used in buildings such as BACnet, Modbus, KNX, and MQTT, plus wireless communication with LoRaWAN, Zigbee, Wi-SUN, and W-MBus devices.

HVAC

Connect, manage, control, and visualize HVAC systems with built-in templates and configurations.

ENERGY MANAGEMENT

Monitor power and energy performance with customizable dashboards and reports.

LIGHTING AND SHADING CONTROL

Easily integrate lighting and shading applications for intuitive operation with zone-based configuration.

RENEWABLES

Integration of renewable energy sources and loads for demand management.

EV CHARGING POINTS

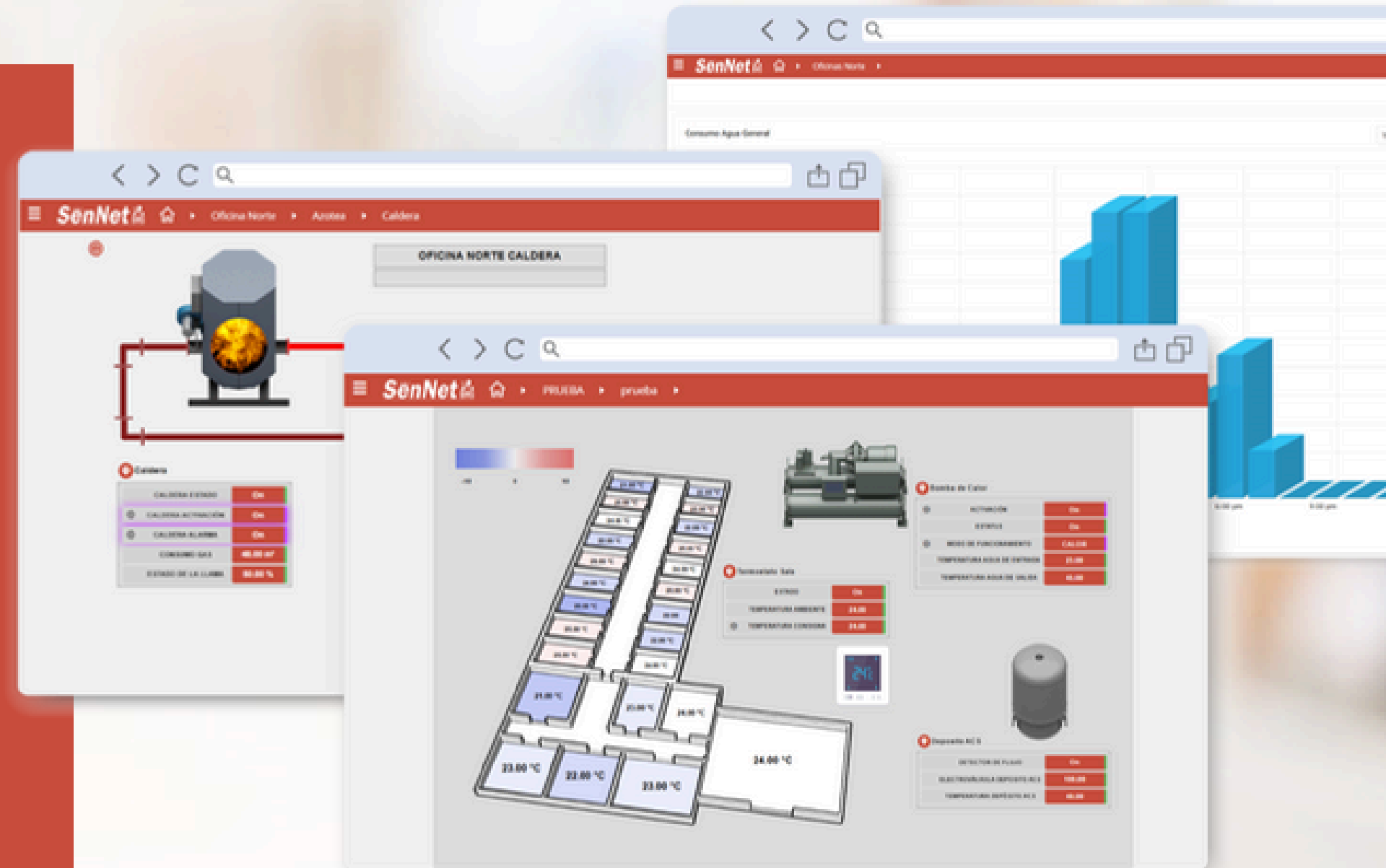
Integrates **OCPP** (Open Charge Point Protocol) to manage EV chargers and define control logic based on schedules, energy cost, photovoltaic surplus, or any rule optimizing the charging network.

SenNet BMS

Edge2Cloud Technology

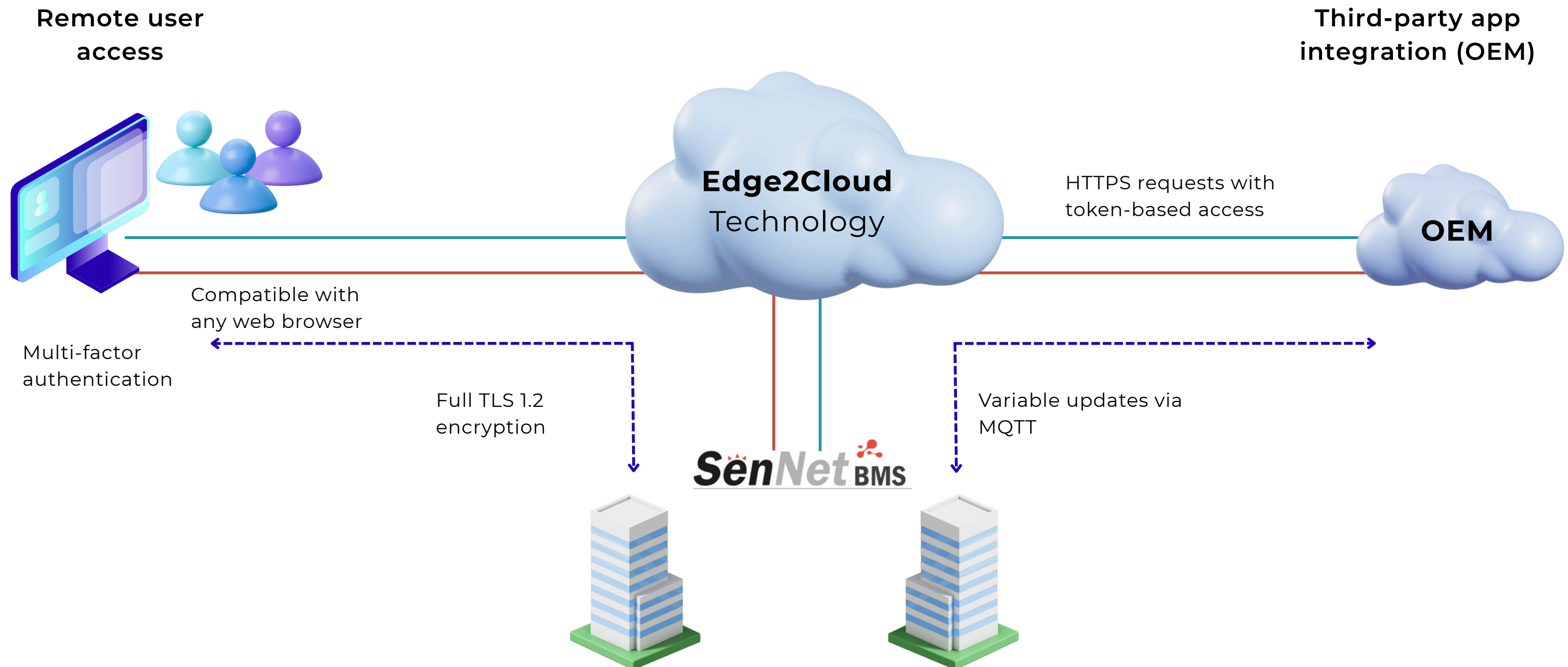
SenNet BMS enables users to access buildings and data easily and securely from anywhere.

- **Open:** Connects and integrates with other cloud platforms via open API.
- **Easy:** Plug & Play deployment across multiple locations.
- **Scalable:** Manage hundreds of sites remotely from any browser.
- **Secure:** Remote connectivity without VPN, using next-generation authentication and encryption.



SenNet BMS

All your buildings accessible from the Cloud



Cybersecure Framework

Guaranteed control security

Cybersecure by design: has successfully passed multiple penetration testing procedures.

Enhanced cybersecurity

Data encryption

Secure password management: mandatory change upon next login.

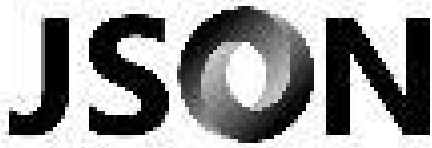
IEC62443 compliance, SAL2 level

Secure web access and navigation



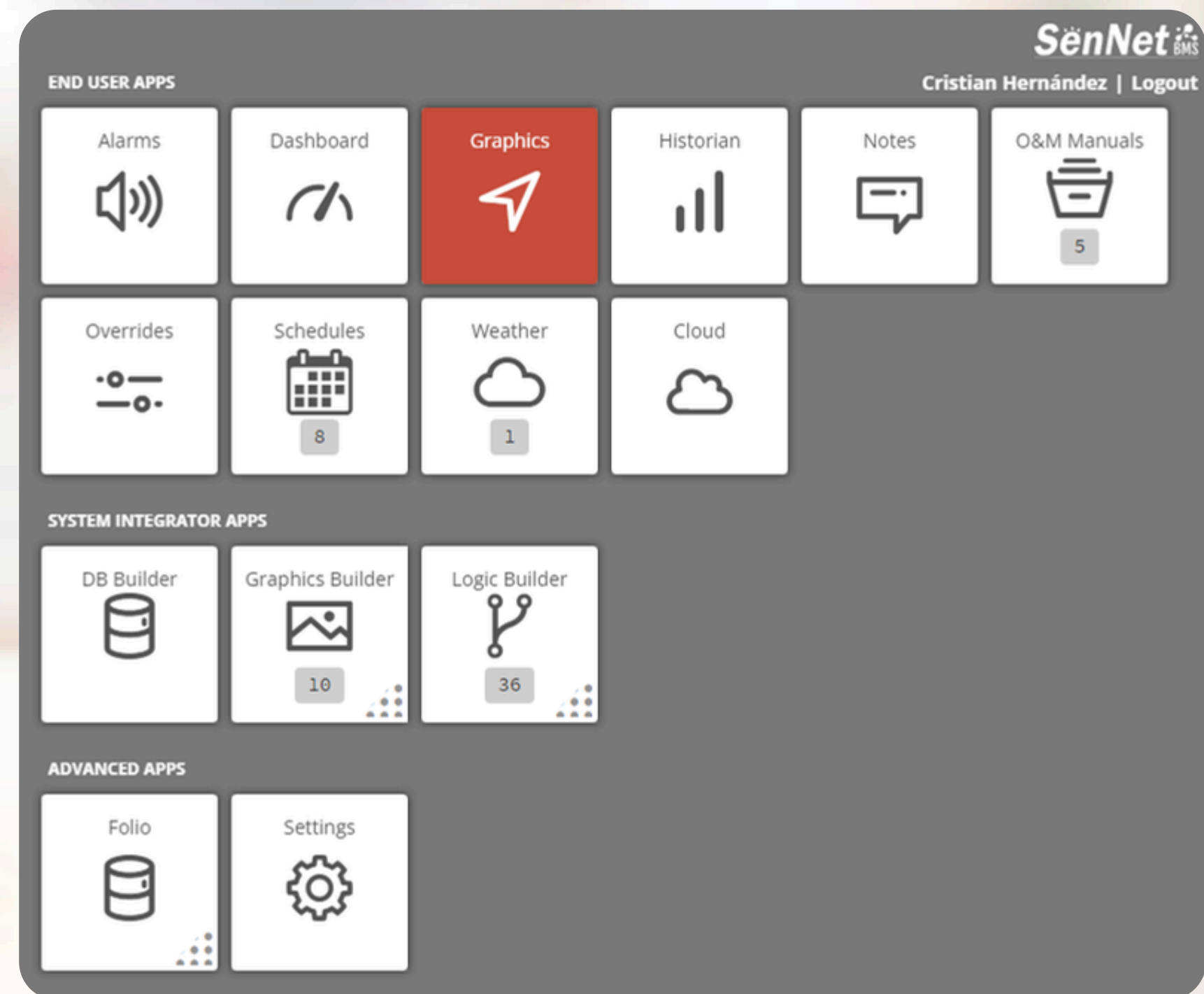
Access to any device within the building

CONNECTORS	
BACnet IP (BTL OWS certified)	SQL
Modbus IP	CSV
Modbus RTU	XML
KNX IP	ASCII file reader
Sedona	Google Calendar
Belimo	JSON
SNMP	MQTT
Obix	Others
nHaystack	



Applications suite

- Alarm management.
- Scheduled reporting.
- Manual change management.
- Calendar and schedule programming.
- Historical data.
- Document management.
- Maintenance notes.
- Interactive dashboards, etc.



Project Haystack

A project to standardize semantic data models and web services.

SenNet BMS leverages Haystack tagging to provide:

- Semantic tagging that unlocks data value, creating real-world benefits.
- Native tagging of project configuration metadata.
- Future-proof hardware/software with an open, widely adopted metadata standard.
- Collaboration with industry groups to expand scope and data types.



Variable tagging

Bringing order and hierarchy to information

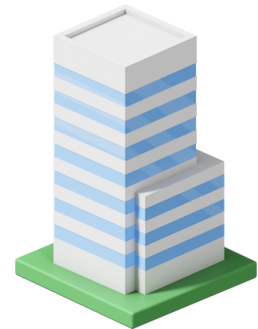


DATA WITHOUT TAGS

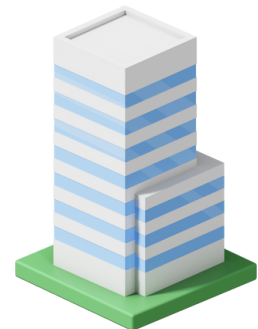
SenNet BMS
APPLIES TAGS

DATA WITH TAGS

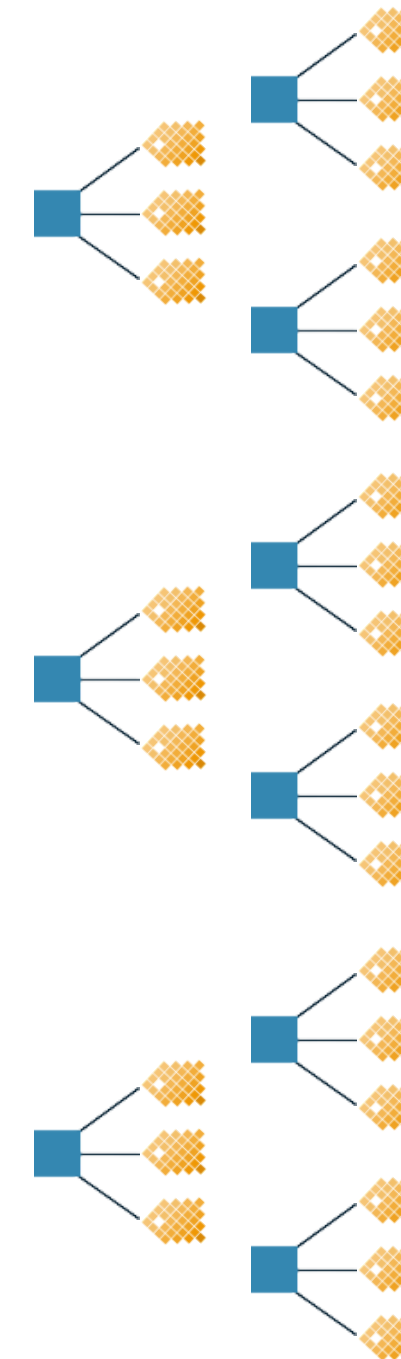
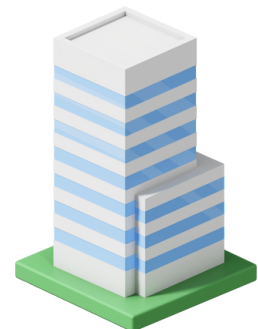
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Data abstraction and
standardization:
independent of their
source.



Mobility

Control buildings and devices from any device.



Open

Integration of any device, easy data transfer to cloud applications.



User Experience

Intuitive interface.



A living solution

Continuous delivery of new, value-added applications.



Data Management

Optimized for data analysis and visualization.



Cybersecurity

Security-centered design, constantly evolving.

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